

NET Centric Computing (CSC378)

Unit 7 - Client-side Development in ASP.NET Core (Theory & Practical)

Updated Practical Version: ASP.NET Core MVC + jQuery

2. Developing Dynamic Web Application using jQuery and ASP.NET Core Web API

Explain the concept of client-side dynamic web development using jQuery and discuss its advantages in modern web applications. Explain how jQuery can communicate with ASP.NET Core Web API using AJAX.

Develop a dynamic web application using jQuery with ASP.NET Core Web API.

Tasks:

1. Create an ASP.NET Core Web API project.
2. Create required API controllers and REST endpoints.
3. Design a client-side interface using HTML, CSS, and jQuery.
4. Implement:
 - jQuery selectors and events.
 - DOM manipulation.
 - Form handling using jQuery.
 - AJAX communication with ASP.NET Core Web API.
 - Dynamic data loading without page refresh.
5. Integrate REST API calls between jQuery client-side code and ASP.NET Core backend.
6. Demonstrate:
 - Sending data from client to server using AJAX POST request.
 - Receiving data from server using AJAX GET request.
 - Displaying dynamic data without refreshing the browser.
7. Document:
 - jQuery-based client-server architecture.
 - jQuery concepts and methods.
 - AJAX request and response process.
 - REST API integration process.
 - Screenshots showing application execution and output.

3. Comparison Study: jQuery vs Modern JavaScript Frameworks

Compare jQuery with modern frontend frameworks for developing interactive web applications with ASP.NET Core backend.

Discuss:

- Architecture.

- Programming approach.
- DOM manipulation.
- Data handling.
- Performance.
- Learning difficulty.
- Advantages and disadvantages.

Provide suitable examples.

4. Practical jQuery Form Development

Create a form application using jQuery.

Tasks:

1. Design an HTML form for user data entry.
2. Implement jQuery functionality for:
 - Selecting form elements.
 - Handling button click events.
 - Reading form input values.
 - Updating webpage content dynamically.
3. Implement client-side validation using jQuery:
 - Required field checking.
 - Email format validation.
 - Password validation.
 - Phone number validation.
4. Handle form submission using jQuery AJAX.
5. Display submitted information dynamically without page reload.

Required Documentation:

- Introduction of client-side development.
- jQuery concepts.
- jQuery selectors and events.
- Forms and validation techniques.
- AJAX communication.
- ASP.NET Core Web API integration.
- Source code.
- Screenshots of implementation and output.